

SRN: PES1UG24CS612

Problem Statement #07 — Hostel Maintenance & Issue Ticketing System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue. Fail: Incomplete form accepted without room tag.	Core function for reporting hostel maintenance issues.	
FR-002	Functional	The system shall allow the Maintenance Warden to assign maintenance tickets to appropriate maintenance staff.	High	Pass: A selected ticket is assigned to a maintenance staff member and the assignment is recorded. Fail: Ticket remains unassigned after a staff member is selected.	Ensures reported issues are dispatched to the responsible staff.	
FR-003	Functional	The system shall allow maintenance staff to update the status of assigned tickets.	High	Pass: Staff can change a ticket status to an available status such as In Progress or Resolved, and the new status is displayed. Fail: Status change is not recorded.	Allows residents and wardens to track maintenance progress.	
FR-004	Functional	The system shall track the SLA timeline of each maintenance ticket.	High	Pass: The system records the ticket creation time and displays whether the ticket is within or beyond its SLA. Fail: SLA status is not updated according to the recorded time.	Helps the administration monitor delayed maintenance work.	
FR-005	Functional	The system shall allow residents to provide a closure sign-off for a resolved maintenance ticket.	Medium	Pass: A resident can confirm closure of a resolved ticket and the sign-off is recorded. Fail: The ticket is marked closed without recording the resident's sign-off.	Confirms that the reported issue has been satisfactorily addressed.	
NFR-001	Nonfunc. (Performance & Security)	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures high-priority tickets are escalated when the SLA is exceeded.	
NFR-002	Nonfunc.	The system shall protect maintenance ticket information so that only authorized users can access or modify ticket records.	High	Pass: Unauthorized users are denied access to ticket records, while authorized users can access permitted functions. Fail: An unauthorized user can view or modify ticket information.	Protects resident and maintenance information from unauthorized access.	